

DENIAL OF SERVICE LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
Instructor:	Ray Lampano, Jnr.
Section Number:	15480/15481
Student Name:	Donaven Bruce
Student ID:	[Redacted]
Submission Platform:	Canvas
Report Date:	2026-01-14
Report Version:	1.0
Lab Title:	Denial of Service
Module Reference:	Denial of Service
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-01-14
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Server), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Workstation), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation). - The workflow centered on PLABWIN10 as the target system, PLABKALI01 as the Linux attack host, and PLABDM01 as an additional Windows-based source for oversized ping traffic. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The module focused on denial-of-service concepts through Wireshark, hping3, Windows Firewall, ICMP flooding, oversized ping traffic generation, and Metasploit SYN flood activity. - Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations. - Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline. - The module aligned most closely with exam objectives 3.2 (Information Security Attack Detection) and 3.3 (Information Security Attack Prevention).	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No production infrastructure, third-party systems, or live services were contacted. - The actions documented here were performed for defensive understanding, traffic visibility, and ethical hacking education within the course environment. - Restricted material and lab credentials are omitted or redacted from this report.	
Work Objectives:	
- Install Wireshark and capture live traffic on the target workstation. - Observe how SYN flooding appears at the packet level and how it affects target responsiveness. - Disable Windows Defender Firewall on PLABWIN10 to permit later ICMP-based testing. - Perform an ICMP flood and observe captured traffic on the target. - Generate oversized ping traffic from a second Windows host and observe elevated network load. - Use the Metasploit synflood module to demonstrate another DoS workflow and review defensive prevention methods for DoS and DDoS activity.	

TECHNICAL ANALYSIS

Wireshark Installation and Initial Packet Capture:

- On PLABWIN10, Internet Explorer was opened and the internal Tools > Hacking Tools area was used to download Wireshark-win32-1.12.3.exe.
- The installer was launched with default component selections, desktop icon enabled, and the bundled WinPcap 4.1.3 dependency installed during setup.
- After installation completed, Wireshark launched successfully and the software update prompt was deferred.
- The Ethernet interface was selected and live packet capture was started on PLABWIN10.
- This established the baseline visibility needed to observe later flood traffic directed at the target workstation.

SYN Flooding with hping3 from PLABKALI01:

- On PLABKALI01, a terminal session was opened and the following command was used:
hping3 -S 192.168.0.3 -a 192.168.0.4 -p 22 -flood
- This command targeted PLABWIN10 at 192.168.0.3, spoofed the apparent source as 192.168.0.4, used TCP SYN packets, targeted port 22, and enabled continuous flooding.
- Back on PLABWIN10, Wireshark displayed the flood of SYN packets and the workstation exhibited resource strain and reduced responsiveness during the capture.
- The attack was then manually stopped from PLABKALI01 with Ctrl + C.
- This stage showed how overwhelming connection-initiation traffic can exhaust resources even before a full session is established.

Windows Defender Firewall Disablement on PLABWIN10:

- On PLABWIN10, Windows Defender Firewall settings were opened through the system search box.
- Firewall protection was turned off for Domain, Private, and Public network profiles as directed by the lab instructions.
- The change was verified in the firewall control panel before the window was closed.
- This was a deliberate lab-only configuration change to allow later traffic demonstrations that would otherwise be filtered by the host firewall.

ICMP Flood Demonstration:

- Wireshark was reopened on PLABWIN10 and packet capture was restarted on the Ethernet interface.
- On PLABKALI01, the terminal was cleared and the following command was used:
hping3 -c 100 -icmp 192.168.0.3
- This command sent 100 ICMP packets to the target host at 192.168.0.3.
- Wireshark on PLABWIN10 showed the corresponding ICMP activity in the packet capture window.
- The output on PLABKALI01 confirmed successful transmission of the packet set.
- This stage contrasted protocol-specific packet flooding with the earlier TCP SYN flood while still demonstrating abnormal traffic concentration on the target.

ATTACK OBSERVATION AND RESOURCE EFFECTS

Oversized Ping Traffic from PLABDM01:

- On PLABDM01, a command prompt was opened from the Run dialog box.
- The following command was used to generate continuous oversized ping traffic:
ping 192.168.0.3 -t -l 65500
- This command sent continuous ping traffic to PLABWIN10, with -t causing indefinite transmission until stopped and -l 65500 specifying the large payload size.
- On PLABWIN10, Task Manager was opened and the Performance tab was used to observe Ethernet activity while the traffic was in progress.
- Elevated network utilization was visible during the traffic generation period.
- After the ping stream was terminated on PLABDM01 with Ctrl + C, the target workstation's network activity dropped back down noticeably.
- This stage emphasized practical system impact observation rather than only packet capture.

Metasploit SYN Flood Module Demonstration:

- On PLABKALI01, the Metasploit Framework was opened and the console was started with:
msfconsole
- Inside Metasploit, the SYN flood auxiliary module was selected with:
use auxiliary/dos/tcp/synflood
- The target host was configured with:
set RHOST 192.168.0.3
- The target port was configured with:
set RPORT 21
- The spoofed source host was configured with:
set SHOST 192.168.0.10
- The timeout value was configured with:
set TIMEOUT 50000
- The module was launched with:
exploit
- While the module ran, PLABWIN10 Task Manager again showed elevated Ethernet activity under the Performance tab.
- The Metasploit-driven flood was manually terminated on PLABKALI01 with Ctrl + C.
- This portion demonstrated a framework-based method of generating denial-of-service traffic rather than relying only on direct packet-generation utilities.

Key Commands, Tools, and Artifacts by System:

PLABWIN10 - Target / Visibility / Performance Observation
Wireshark-win32-1.12.3.exe installed with WinPcap 4.1.3
Ethernet interface selected for packet capture
Windows Defender Firewall turned off for Domain, Private, and Public profiles
Task Manager > Performance > Ethernet used to observe traffic spikes

PLABKALI01 - Attack Generation / Framework Use
hping3 -S 192.168.0.3 -a 192.168.0.4 -p 22 -flood
hping3 -c 100 -icmp 192.168.0.3
msfconsole
use auxiliary/dos/tcp/synflood
set RHOST 192.168.0.3
set RPORT 21
set SHOST 192.168.0.10
set TIMEOUT 50000
exploit

PLABDM01 - Additional Traffic Source
ping 192.168.0.3 -t -l 65500

Primary evidence and artifacts observed
Wireshark capture screens, hping3 execution output, firewall status confirmation,
Task Manager Ethernet activity graphs, Metasploit synflood console output.

Operational Notes and Lab Behavior:

- The lab specifically noted that PLABWIN10 could become slow, unstable, or temporarily unresponsive during flood activity due to resource exhaustion.
- The out-of-memory condition mentioned by the vendor materials was environment-dependent and not guaranteed on every run.
- Because the module used deprecated software and a legacy lab image, observed behavior was best treated as a controlled demonstration of impact rather than a modern resilience benchmark.
- The lab materials used the phrase "Ping of Death," but the demonstrated workflow was a Windows continuous large-payload ping used to show elevated traffic and performance impact in the sandbox.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered packet capture, SYN flooding, firewall modification, ICMP flooding, oversized ping traffic generation, and Metasploit-based SYN flood activity.
- The recurring theme across the module was that denial-of-service activity can be studied through both packet-level evidence and system-level performance symptoms.
- The prevention portion reinforced that visibility alone is not enough; filtering, throttling, capacity planning, proxying, and service hardening all contribute to resilience.

MITRE ATT&CK Mapping:

- Impact: T1498 Network Denial of Service - the lab's SYN and ICMP flooding activities aligned most directly with network-level service degradation through traffic saturation.
- Impact: T1499 Endpoint Denial of Service - target-side resource exhaustion and degraded host responsiveness on PLABWIN10 reflected endpoint-focused availability impact.
- Defense Evasion / Impair Defenses context: disabling Windows Defender Firewall reduced a host protection layer and demonstrated how defensive controls change attack observability and exposure.
- Detection context: packet capture and performance monitoring illustrated how defenders can identify DoS conditions through both traffic signatures and abnormal resource usage.

Detection Opportunities:

- Packet capture tools such as Wireshark can reveal repeated SYN or ICMP patterns inconsistent with normal traffic baselines.
- System-side indicators such as lag, UI unresponsiveness, and unusually high network utilization provide visible signs of service-degradation attempts.
- Traffic baselining, anomaly detection, and profiling of expected bandwidth consumption can help distinguish legitimate bursts from hostile flooding.
- Sequential change-point methods and volume-based anomaly analysis are useful when sudden traffic deviations need to be flagged quickly.

Defensive Controls:

- Use firewalls to filter and rate-limit hostile incoming traffic where appropriate.
- Maintain endpoint anti-malware and system hardening to reduce exposure to attack tooling and secondary compromise.
- Provision adequate bandwidth and use router throttling or upstream mitigation where available.
- Use WAF protection for web-facing applications and reverse proxies to absorb or distribute load.
- Disable unnecessary services, monitor servers regularly, and audit abnormal spikes in traffic or resource consumption.
- Filter unwanted e-mail and reduce exposure to botnet-enablement or indirect DDoS preparation paths.

Completion Verification:

- Work was completed across Wireshark installation, SYN flood observation, firewall disablement, ICMP flood testing, oversized ping traffic observation, and Metasploit synflood execution.
- Completion was evidenced by visible Wireshark flood traffic, hping3 console output, confirmed firewall status changes, Task Manager Ethernet spikes, and active Metasploit module execution.
- The report therefore documents both the attack-demonstration workflow and the prevention-oriented lessons tied to DoS/DDoS detection and mitigation.

Key Findings:

- 1) SYN flooding can be identified clearly in packet capture data and can quickly degrade endpoint responsiveness when traffic is sustained.
- 2) ICMP flooding differs at the protocol level but produces the same broader defensive problem: abnormal concentration of traffic against a target.
- 3) Task Manager performance metrics complemented packet capture by showing resource impact in a way that is immediately visible to administrators.
- 4) Firewall state materially affects exposure, especially for packet types that would otherwise be restricted at the host boundary.
- 5) The most valuable defensive takeaway was that DoS resilience depends on layered controls rather than a single detection method.

Issues / Deviations / Notes:

- The lab materials used the phrase "SYN Flood" in one section, but the demonstrated activity was clearly a SYN flood attack.
- Because the platform used deprecated tools and a legacy environment, exact performance impact may vary across runs or differ from modern production systems.
- The prevention section was conceptual rather than configuration-heavy, so the strongest value of the module was in linking traffic symptoms to higher-level mitigation concepts.
- The actual commands used in the lab are now preserved directly in this report for reproducibility and accurate documentation of the completed workflow.

Reflection:

- This work tied together packet-level evidence, host-level performance effects, and prevention concepts in a way that made denial-of-service behavior easier to recognize and explain.
- The strongest takeaway was that service degradation is often visible in multiple places at once: inside packet captures, inside system performance tools, and inside user-facing instability.
- The completed workflow supports future ethical hacking and defensive analysis by showing how flood traffic can be generated, detected, observed, and mapped to layered defensive controls.